Constant Speed

$d = vt$

Constant Acceleration

$v = v_0 + at$

$v^2 = v_0^2 + 2a(x - x_0)$

$x = x_0 + v_0 t + \frac{1}{2}at^2$

$x = x_0 + \frac{1}{2}(v_0 + v)t$

$x = x_0 + vt - \frac{1}{2}at^2$

Projectile Motion

$x_f = x_0 + (v_0 cos\theta)t$

$y_f = y_0 + (v_0 sin\theta)t$

$v_y = v_0 sin\theta - gt$

$v_x = v_0 cos\theta$

$v_y = (v_0 sin\theta)^2 - 2g(y_f - y_0)$

Trajectory - (When no time)

$y = xtan\theta - \frac{1}{2}g\frac{x^2}{2(v_0 cos\theta)^2}$

Range, Max Height, etc

$R_{Horizontal\ Range} = \frac{v_0^2}{g}sin2\theta$

Max Height = $\frac{v^2 sin^2\theta}{2g}$

$t_{max\ height} = \frac{v_0 sin\theta}{g}$

$t_{time\ in\ air} = \frac{2v_0 sin\theta}{g}$

Circular Motion

$a_c = \frac{v^2}{R}, \ F_c = m(\frac{v^2}{R})$

$T_{this\ is\ period\ btw} = \frac{2\Pi r}{v}$

Rotational Kinematics

$\theta = \omega t$, θ = Angular Position

$\omega = \omega_0 + \alpha t$, ω = Ang. Velocity

$\theta - \theta_0 = \omega_o t + \frac{1}{2}\alpha t^2$, α = Ang. Acceleration

$\tau_{this\ is\ torque} = I\alpha = |F|rsin\theta$

Friction

$F_k = \mu_k N$

$F_s \geq \mu_s N$

Relative Velocity

$v_{BA} = v_B - v_A$

Magnitude of Velocity

$|v| = \sqrt{v_x^2 + v_y^2}$

$v_x = vcos\theta$, $v_y = vsin\theta + at$

Work

$W = |\vec{F}||\vec{d}|cos\theta$

Kinetic Energy

$KE = \frac{1}{2}mv^2$

Work Energy Theorem

$W_{net} = \Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_0^2$

Work done by gravity

$W_g = mgdcos\theta$

Work done when lifting/lowering object

$W_{applied} = -W_g$

Hooke's law

$F_{spring} = -kx$

Potential Energy in Spring: $PE_{spring} = \frac{1}{2}kx^2$

Kinetic Energy in Spring: $KE = \frac{1}{2}mv^2$ (just da same)

Work done by Spring

$W_s = -\frac{1}{2}kx^2$, IF $x_0 \neq 0$, you do:

$W_s = \frac{1}{2}kx_0^2 - \frac{1}{2}kx_f^2$

Different Formulas for Potential Energy:

$\Delta PE = -W$

$PE_{gravitational} = mgh = mg(\Delta y)$ (USE THIS ONE MAINLY)

Elastic Potential Energy

$PE = \frac{1}{2}kx^2$

Mechanical Energy

$ME = KE + PE$

Conservation of Energy:

$KE_o + PE_0 = KE_f + PE_f$, which just equals...

$\frac{1}{2}mv_0^2 + mgh_0 = \frac{1}{2}mv_f^2 + mgh_f$

A more accurate Consv. of Energy:

$W + PE_0 + KE_0 = PE_f + KE_f + E_{lost}$

Work on System by External Force:

$W = \Delta KE + \Delta PE + \Delta E_{lost}$

Center of Mass:

$x_{com} = \frac{\sum_{i=1}^{N} m_i x_i}{M_{Total\ Mass}}$

Linear Momentum:

$\vec{p} = m\vec{v}$

Impulse Momentum Theorem

$\Delta\vec{p} = \vec{p_f} - \vec{p_0} = J$

Conservation linear momentum

$p_0 = p_f$

Elastic Collision (when objects collide and DON'T stay together)

$p_{1i} + p_{2i} = p_{1f} + p_{2f}$

$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$

Inelastic Collision (When objects collide and stick together)

$m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2)v_f$

Impulse: $I = F\Delta t = \Delta p = m\Delta v$

Extra Equations for Work:

$W = F_x d_x + F_y d_y$

Rotational Motion:

$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$

$\omega = \omega_0 + \alpha t$

$\omega^2 = \omega_0^2 + 2\alpha\theta$

$v = \omega r$, $a = \alpha R$, $\theta = \omega t$, $S = \theta R$ [arc length]

$a_c = \omega^2 R$, [radial acceleration]

Torque:

$T = F * R = FRsin\theta = I\alpha$

Rotational Kinetic Energy: (NOT FULL PART THO SEE Rolling without slipping KE for total kinetic energy)

$KE_{rotation} = \frac{1}{2}I\omega^2$, I here is inertia (will vary)

If you have a problem with a wheel moving with friction applied, then you can also do:

$\tau = I\alpha \rightarrow F_f * R = I(\frac{a}{R})$ (you can use this to find acceleration)

Smooth rolling wheel has kinetic energy:

$KE = \frac{1}{2}I_{com}\omega^2 + \frac{1}{2}Mv^2_{com}$

If wheel roll down ramp smoothly, it's acceleration along x-axis extending up the ramp is:

$a_{com,x} = -\frac{gsin\theta}{(\frac{I+I_{com}}{MR^2})}$

Rolling without slipping KE:

$KE_{rolling} = \frac{Iv^2}{2R^2} + \frac{1}{2}Mv^2 = \frac{1}{2}I\omega^2 + \frac{1}{2}Mv^2$ (basically $KE_{rotation} + KE_{translation}$)

Angular Momentum: $\iota = I\omega = mvR = pR$

$\vec{\iota} = \vec{r} * \vec{p} = m(\vec{r} * \vec{v})$

The magnitude $\iota = rmvsin\phi = rp_\perp = rmv_\perp$

ϕ is the angle between $\vec{r}$ *and* $\vec{p}$

IF $\tau_{net} = 0$, *then* $\Delta\iota = 0$, *meaning angular momentum is conserved:*

$I_0\omega_0 = I_f\omega_f$

$W = \tau\theta$

Gravitation:

Any particle in the universe attracts any other particle with a gravitational force whose magnitude is

$F_g = G\frac{m_1 m_2}{r^2}$, $G = 6.67 * 10^{-11}$

Gravitation Near Earth's Surface

$F = ma_g$

$a_g = \frac{GM}{r^2}$

Gravitation Inside Earth

$M_{inside} = \frac{4}{3}\Pi r^3\rho = \frac{M}{R^3}r^3$

^^ρ is the solid spheres density, R is its radius, M is its mass, r is the distance from the center.

Magnitude of force acting on mass m is

$F = \frac{GmM}{R^3}r$

Gravitational Potential Energy:

$U = -\frac{GMm}{r}$

Ex when system contains 3 objects instead of 2:

$U = -(\frac{Gm_1m_2}{r_{12}} + \frac{Gm_1m_3}{r_{13}} + \frac{Gm_2m_3}{r_{23}})$

$v = \sqrt{\frac{2GM}{R}}$ (escape speed from gravitational pull of astronomical body mass M)

Law of Periods:

$T^2 = (\frac{4\Pi^2}{GM})r^3$

M is the mass of the attracting body-the sun in the case of solar system

When a planet/satellite with mass m moves in a circular orbit with radius r, its PE (U), and KE are given by:

$U = -\frac{GMm}{r}$ and $KE = \frac{GMm}{2r}$

The mechanical energy is then: $ME = -\frac{GMm}{2r}$

Density:

$\rho = \frac{\Delta m}{\Delta V}$ or $\rho = \frac{m}{V}$

Pressure:

$p = \frac{\Delta F}{\Delta A}$ or $p = \frac{F}{A}$ (units for pressure is Pa)

$p = \frac{\rho V g}{A}$

The Force resulting from fluid pressure at a particular point in a fluid, has the same magnitude in all directions

$1\ ATM = 1.01 \times 10^5 = 760\ torr = 14.7\frac{lb}{in^2}$

Fluids at rest:

Pressure in a fluid at rest varies with vertical position y. For y measured positive upward,

$p_2 = p_1 + \rho g(y_1 - y_2)$

If h is the depth of a fluid sample below some reference level at which the pressure is p_0, this equation can just become:

$p = p_0 + \rho gh$

Pascals Principle: *states that a change in the pressure applied to an enclosed fluid is transmitted undiminished to every portion of the fluid and to the walls of the containing vessel.*

$p_s = p_L \rightarrow \frac{F_1}{A_1} = \frac{F_2}{A_2}$

Archimedes Principle: *states that when a body is fully or partially submerged in a fluid, the fluid pushes upward with a buoyant force:*

$F_{buoyancy} = m_f g$

The flow within any tube of flow obeys the equation of continuity:

$R_v = Av$

In which R_v is the volume flow rate, A is the area of flow at any point, V is the speed of the fluid at that point.

The Mass flow rate is:

$R_m = \rho R_v = \rho Av$

$A_1V_1 = A_2V_2 \rightarrow$ This is only when the fluid is the same at any point

Bernoulli's equation: applying the principle of conservation of mechanical energy to the flow of an ideal fluid leads to Bernoulli's equation:

$p + \frac{1}{2}\rho v^2 + \rho gh \rightarrow$ *dont confuse the p and ρ, p is pressure, ρ is density*

$p_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = p_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2$

Oscillations:

$T = \frac{1}{f}$, $f = \frac{1}{T}$ T is the time required for one complete cycle (**period**), f is the number of oscillations per second (**frequency**)

In SHM, the displacement x(t) of a particle from its equilibrium position is:

$x = x_m cos(\omega t + \phi)$ x_m is the amplitude

The **angular frequency** ω is related to the period and frequency of the motion by $\omega = \frac{2\Pi}{T} = 2\Pi f$

$v = -\omega x_m sin(\omega t + \phi)$ (velocity)

$a = -\omega^2 x_m cos(\omega t + \phi)$ (acceleration)

Something with mass m that moves under the influence of a Hookes law restoring force given by F = -kx is a linear simple harmonic oscillator with:

$\omega = \sqrt{\frac{k}{m}}$ (angular frequency)

$T = 2\Pi\sqrt{\frac{m}{k}}$ (period)

$\frac{f_2}{f_1} = \sqrt{\frac{k_2}{k_1}}$ *(only use this when you're given two different k values,)*

$k = m\omega^2$ (when force applied)

Energy in Simple Harmonic Motion (SHM)

- A particle in SHM has at any time, kinetic energy $KE = \frac{1}{2}mv^2$ and potential energy $PE = \frac{1}{2}kx^2$

If no friction present then $ME = KE + PE$

here U is just PE

$U(t) = \frac{1}{2}kx^2 = \frac{1}{2}kx_m^2 cos^2(\omega t + \phi)$

$KE(t) = \frac{1}{2}mv^2 = \frac{1}{2}kx_m^2 sin^2(\omega t + \phi)$

$ME = \frac{1}{2}kx_m^2$

$a_{max} = \frac{kA}{m} = A\omega^2$ where A is amplitude, so $A = x_m$

$v_{max} = A\sqrt{\frac{k}{m}}$

Simple Pendulum:

$T = 2\Pi\sqrt{\frac{I}{mgL}}$ (simple pendulum)

I is inertia, m is mass, L is rods length

Physical Pendulum:

$T = 2\Pi\sqrt{\frac{I}{mgh}}$ (physical pendulum)

I is inertia, m is pendulums mass, h is distance between pivot and pendulums center of mass.

Pendulum by Organic Chem Tutor on youtube:

$T = 2\Pi\sqrt{\frac{L}{g}}$

$L = \frac{gT^2}{4\Pi^2}$

Temperature:

$T = (273.16)(\lim_{gas \rightarrow 0} \frac{p}{p_3})$

The Celsius temperature scale is defined by : $T_c = T - 273.15^o$

Fahrenheit: $T_F = \frac{9}{5}T_c + 32^o$

Thermal Expansion:

For a temperature change ΔT, a change ΔL in any linear dimension is given by

$\Delta L = \alpha L_0 \Delta T$ α here is the coefficient of linear expansion

The change ΔV in the volume V of a solid or liquid is

$\Delta V = V_0\beta\Delta T$ β here is the coefficient of volume expansion, also $\beta = 3\alpha$

$\Delta A = L_0 W_0[2\alpha\Delta T + \alpha^2\Delta T^2]$ A is area, L is length, W is width

$\Delta A = 2L_0 W_0 \alpha\Delta T$ You can rewrite the area like this cause its almost close